
MODULE *tcommit*

This specification is explained in “Transaction Commit”, Lecture 5 of the TLA+ Video Course.

CONSTANT *RM* The set of participating resource managers

VARIABLE *rmState* *rmState*[*rm*] is the state of resource manager *rm*.

TCTypeOK $\triangleq$

The type-correctness invariant

$rmState \in [RM \rightarrow \{\text{“working”}, \text{“prepared”}, \text{“committed”}, \text{“aborted”}\}]$

TCInit $\triangleq$ $rmState = [r \in RM \mapsto \text{“working”}]$

The initial predicate.

canCommit $\triangleq$ $\forall r \in RM : rmState[r] \in \{\text{“prepared”}, \text{“committed”}\}$

True iff all *RM*s are in the “prepared” or “committed” state.

notCommitted $\triangleq$ $\forall r \in RM : rmState[r] \neq \text{“committed”}$

True iff no resource manager has decided to commit.

We now define the actions that may be performed by the *RM*s, and then define the complete next-state action of the specification to be the disjunction of the possible *RM* actions.

Prepare(*r*) $\triangleq$ $\wedge rmState[r] = \text{“working”}$
 $\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{“prepared”}]$

Decide(*r*) $\triangleq$ $\vee \wedge rmState[r] = \text{“prepared”}$
 $\wedge canCommit$
 $\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{“committed”}]$
 $\vee \wedge rmState[r] \in \{\text{“working”}, \text{“prepared”}\}$
 $\wedge notCommitted$
 $\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{“aborted”}]$

TCNext $\triangleq$ $\exists r \in RM : Prepare(r) \vee Decide(r)$

The next-state action.

TCConsistent $\triangleq$

A state predicate asserting that two *RM*s have not arrived at conflicting decisions. It is an invariant of the specification.

$\forall r1, r2 \in RM : \neg \wedge rmState[r1] = \text{“aborted”}$
 $\wedge rmState[r2] = \text{“committed”}$

The following part of the spec is not discussed in Video Lecture 5. It will be explained in Video Lecture 8.

TCSpec $\triangleq$ $TCInit \wedge \Box [TCNext]_{rmState}$

The complete specification of the protocol written as a temporal formula.

THEOREM $TCSpec \Rightarrow \Box(TCTypeOK \wedge TCConsistent)$

This theorem asserts the truth of the temporal formula whose meaning is that the state predicate $TCTypeOK \wedge TCInvariant$ is an invariant of the specification $TCSpec$. Invariance of this conjunction is equivalent to invariance of both of the formulas $TCTypeOK$ and $TCConsistent$.

\ * Modification History
\ * Last modified Sun Dec 07 11:17:10 EET 2025 by *zygis*
\ * Created Sun Dec 07 11:16:53 EET 2025 by *zygis*